Last Time

Populations vs. Samples (quick)

Statistics and Sample Spaces

Distributions

Quantiles

Distribution Shapes

Central Tendency

Today

Central Tendency

Properties of Central Tendency Measures

Summation (Sigma) Notation

Summation Rules

The Mean as a Geometric Center

Dispersion

Measures of Central Tendency

The middle or most characteristic score; a measurement of the dataset X.

- mean: $\overline{X} = \frac{1}{N} \sum_{i=1}^{N} X_i$.
- median: If $Pr(X \leq x) \geq .5$ and $Pr(X \geq x) \geq .5$, then $Md = x$
- mode: If $x = \text{argmax}_{x_i} Pr(X = x_i)$, then $Mo = x$.

GSS "Health" Responses

Relative Frequency	Health Rating 1	2	3	4
$p_X(x)$	0.37	0.46	0.15	0.03
$P_X(x)$	0.37	0.83	0.98	1.00

- mean
- median
- mode

Weighted Means and Expected Values:

$$\overline{X} = \sum_{i=1}^{m} x_i p_X(x_i)$$

Weighted Means

```
> # Central tendency
> library('DescTools') # Has some useful stuff
> mean(data$health)
[1] 1.836197
> median(data$health)
[1] 2
> Mode(data$health) # Mode function from DescTools package)
[1] 2
attr(,"freq")
[1] 400
>
> # Weighted mean
> p.Health <- table(data$health)
> Health <- as.numeric(names(p.Health))
```

Properties of Central Tendency Measures

Shoe Sizes

$X = \{11, 11.5, 5.5, 8, 10, 6, 7.5, 9.5, 10, 6\}$

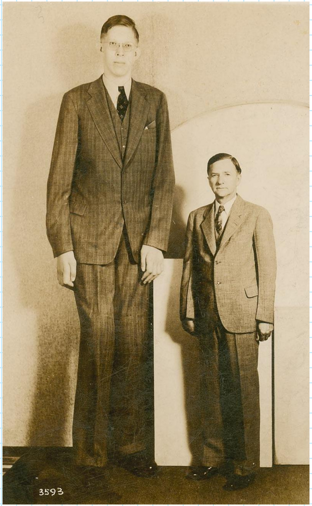
- Mode
- Median
- Mean

Properties of Central Tendency Measures

- Mode
  - Only appropriate measure of central tendency for categorical/nominal data
  - Unstable
  - Not necessarily unique
- Median
  - Appropriate for ordinal or higher scales of measurement
  - Robust to outliers
  - Not necessarily unique
  - Minimizes the Mean Absolute Deviation (MAD)
- Mean
  - Appropriate for interval or higher scales of measurement
  - Very sensitive to outliers
  - Is the unique centroid/geometric center/balance point of a distribution
  - Minimizes the Sum of Squared Deviations

Sensitivity to Outliers

Robert Pershing Wadlow (1918-1940)
(Let’s go back!)



Deviation Scores

$$D_i = X_i - \overline{X}$$

- What is $\sum_{i=1}^{N} D_i$?
- What is $\sum_{i=1}^{N} D_i^2$?

Sum of Squares

$$\sum_{i=1}^{N} D_i^2 = \sum_{i=1}^{N} (X_i - \overline{X})^2$$

Summation (Sigma) Notation

Example

$$\overline{X} = \frac{1}{N}\sum_{i=1}^{N} X_i$$

Summation Notation

$$\sum_{i=M}^{N} X_i = X_M + X_{M+1} + \cdots X_N$$

Summation Notation

$$\sum_{i=M}^{N} X_i = X_M + X_{M+1} + \cdots X_N$$

Rewrite the following in summation notation:

$$(X_1 + X_2 + X_3 + X_4)/4 \quad =$$

Summation Notation

$$\sum_{i=M}^{N} X_i = X_M + X_{M+1} + \cdots X_N$$

Rewrite the following in summation notation:

$$X_1 Y_1 + X_2 Y_2 + \cdots + X_N Y_N \quad =$$

Summation Notation

$$\sum_{i=M}^{N} X_i = X_M + X_{M+1} + \cdots X_N$$

Rewrite the following in summation notation:

$$(X_1^2 - 2X_1 + 3) + 2(X_2^2 - 2X_2 + 3) + \cdots + N(X_N^2 - 2X_N + 3) \quad =$$

Summation Notation

$$\sum_{i=M}^{N} X_i = X_M + X_{M+1} + \cdots X_N$$

Rewrite the following in extended form:

$$\sum_{i=5}^{7} X_i \quad =$$

Summation Notation

$$\sum_{i=M}^{N} X_i = X_M + X_{M+1} + \cdots X_N$$

Rewrite the following in extended form:

$$\sum_{i=1}^{6} X_i^2 \quad =$$

Summation Notation

$$\sum_{i=M}^{N} X_i = X_M + X_{M+1} + \cdots X_N$$

Rewrite the following in extended form:

$$\sum_{i=1}^{5} \sum_{j=1}^{4} (X_{ij}^2 + Y_i) \quad =$$

Summation Rules

- *Rule 1: The Sum of a Variable V.*

$$\sum_{i=1}^{n} V_i = V_1 + V_2 + \cdots + V_n.$$

- *Rule 2: The Sum of a Constant.*

$$\sum_{i=1}^{n} c$$

Summation Rules

- *Rule 1: The Sum of a Variable V.*

$$\sum_{i=1}^{n} V_i = V_1 + V_2 + \cdots + V_n.$$

- *Rule 2: The Sum of a Constant.*

$$\sum_{i=1}^{n} c = c + c + \cdots + c = nc.$$

Summation Rules

- *Rule 3: The Sum of a Product of a Constant and a Variable.*

$$\sum_{i=1}^{n} cV_i$$

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$$\sum_{i=1}^{n} cV_i = cV_1 + cV_2 + \cdots + cV_n = c(V_1 + \cdots + V_n) = c\sum_{i=1}^{n} V_i$$

Summation Rules

- *Rule 3: The Sum of a Product of a Constant and a Variable.*

$$\sum_{i=1}^{n} cV_i = cV_1 + cV_2 + \cdots + cV_n = c(V_1 + \cdots + V_n) = c\sum_{i=1}^{n} V_i, \text{ and}$$

$$\sum_{i=1}^{n} \frac{V_i}{c} = \frac{1}{c}\sum_{i=1}^{n} V_i,$$

Summation Rules

- *Rule 4: Distribution of Summation*.

$$\sum_{i=1}^{n}(V_i + W_i) \quad =$$

Summation Rules

- *Rule 4: Distribution of Summation.*

$$\begin{aligned}\sum_{i=1}^{n}(V_i + W_i) &= (V_1 + W_1) + (V_2 + W_2) + \cdots + (V_n + W_n) \\ &= (V_1 + \cdots + V_n) + (W_1 + \cdots + W_n) \\ &= \sum_{i=1}^{n} V_i + \sum_{i=1}^{n} W_i,\end{aligned}$$

Summation Rules

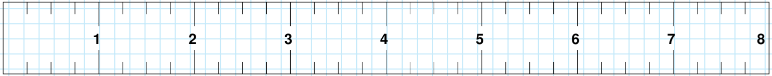
- *Rule 4: Distribution of Summation*.

$$\begin{aligned}
\sum_{i=1}^{n}(V_i + W_i) &= (V_1 + W_1) + (V_2 + W_2) + \cdots + (V_n + W_n) \\
&= (V_1 + \cdots + V_n) + (W_1 + \cdots + W_n) \\
&= \sum_{i=1}^{n} V_i + \sum_{i=1}^{n} W_i, \text{ and}
\end{aligned}$$

$$\sum_{i=1}^{n}(V_i - W_i) = \sum_{i=1}^{n} V_i - \sum_{i=1}^{n} W_i.$$

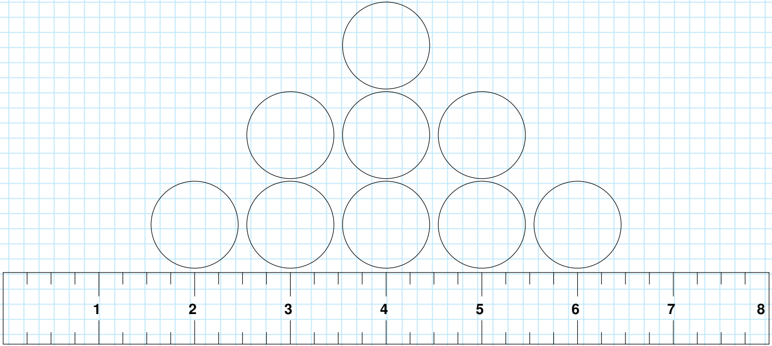
The Balance Point

$$\{2, 3, 3, 4, 4, 4, 5, 5, 6\}$$



The Balance Point

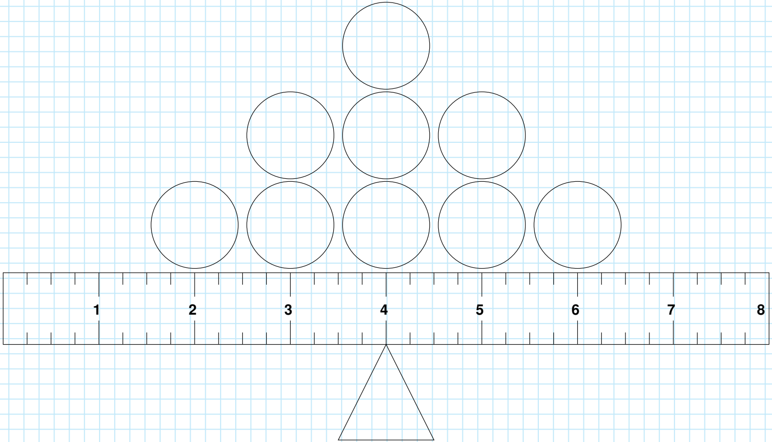
$$\{2, 3, 3, 4, 4, 4, 5, 5, 6\}$$



The Balance Point

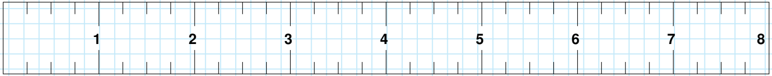
$\{2, 3, 3, 4, 4, 4, 5, 5, 6\}$

$$\overline{X} = \frac{1}{9}\sum_{i=1}^{9} X_i = \frac{1}{9}(2+3+3+\cdots+6) = \frac{36}{9} = 4$$



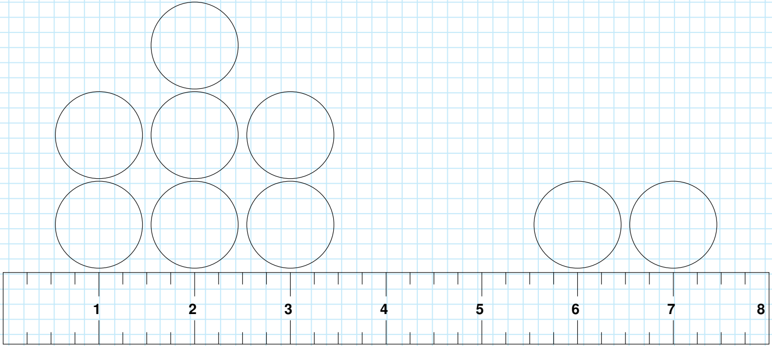
The Balance Point

$$\{1, 1, 2, 2, 2, 3, 3, 6, 7\}$$



The Balance Point

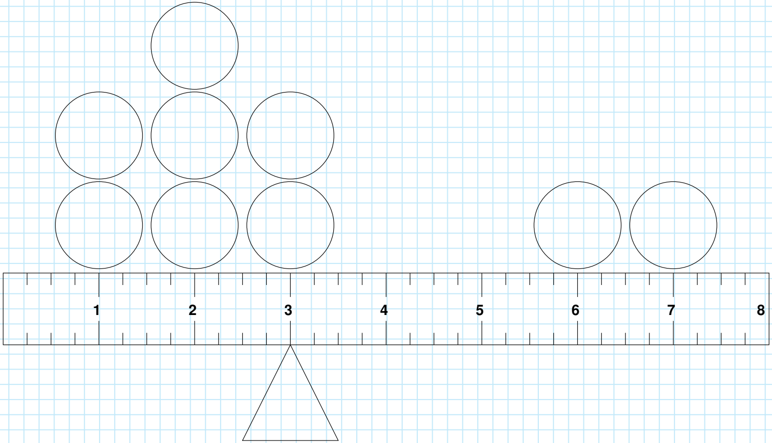
$$\{1, 1, 2, 2, 2, 3, 3, 6, 7\}$$



The Balance Point

$\{1, 1, 2, 2, 2, 3, 3, 6, 7\}$

$$\overline{X} = \frac{1}{9}\sum_{i=1}^{9} X_i = \frac{1}{9}(1 + 1 + 2 + \cdots + 7) = \frac{27}{9} = 3$$



Proof that the Mean is the Balance Point

Definition: For a sample of scores $\{X_1, X_2, \ldots, X_N\}$, the *deviation scores* $\{D_1, D_2, \ldots, D_N\}$ are all the differences between the scores (X) and their mean ($\bar{X}$).

$$D_i = X_i - \overline{X}$$

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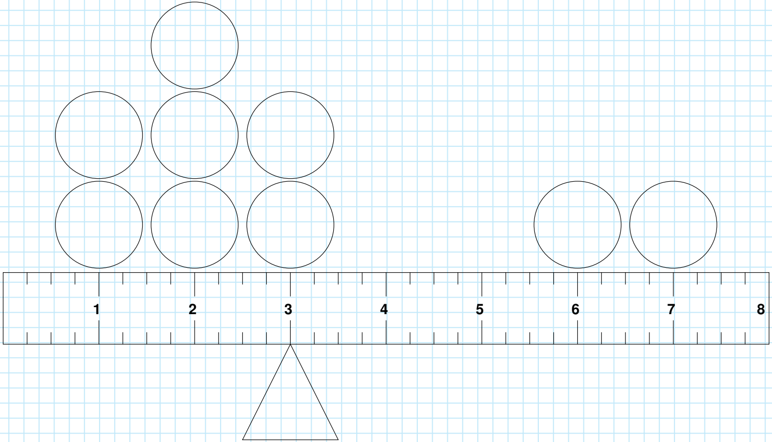
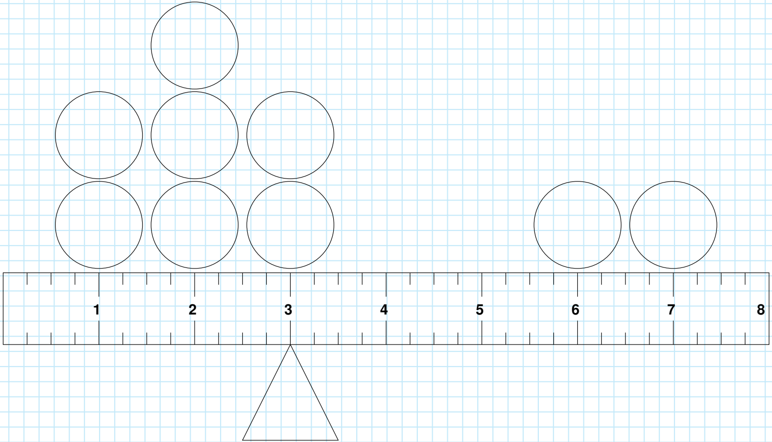


Proof that the Mean is the Balance Point

For the mean to “balance” the scores, for every $D_i < 0$ there must be some postive $D_j s > 0$ that zero it out.



Proof that the Mean is the Balance Point

For the mean to “balance” the scores, for every $D_i < 0$ there must be some $D_j s > 0$ that zero it out. That is,

$$\sum_{i=1}^{N} D_i = 0.$$

Proof

$$\sum_{i=1}^{N} D_i = \sum_{i=1}^{N} (X_i - \overline{X})$$

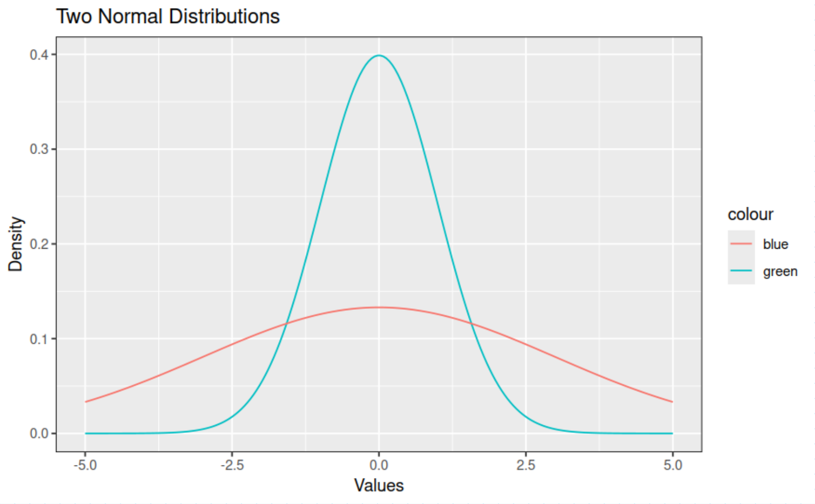
Proof

$$\begin{aligned}
\sum_{i=1}^{N} D_i &= \sum_{i=1}^{N} (X_i - \overline{X}) \\
&= \sum_{i=1}^{N} X_i - \sum_{i=1}^{N} \overline{X} \\
&= \sum_{i=1}^{N} X_i - N\overline{X} \\
&= \sum_{i=1}^{N} X_i - N\frac{1}{N}\sum_{i=1}^{N} X_i \\
&= \sum_{i=1}^{N} X_i - \sum_{i=1}^{N} X_i = 0.
\end{aligned}$$

Dispersion

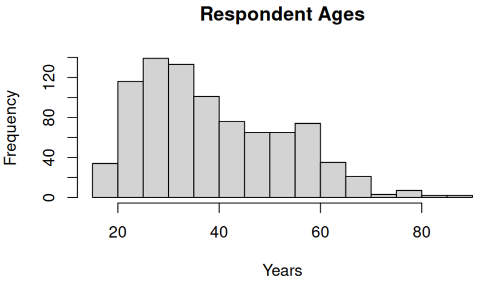
- By how much do the values in the sample deviate from the "most characteristic score"?
- The central tendency is a prediction about the other values in the sample.

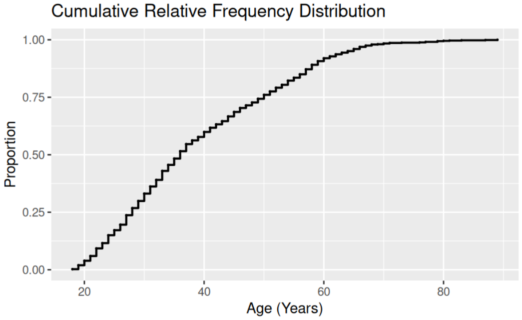
Dispersion



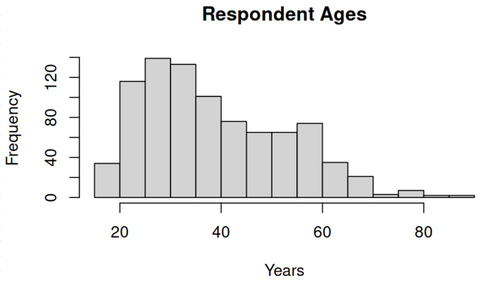
Two Normal Distributions
0.4
0.3
0.2
0.1
0.0
Density
-5.0
-2.5
0.0
2.5
5.0
Values
colour
blue
green

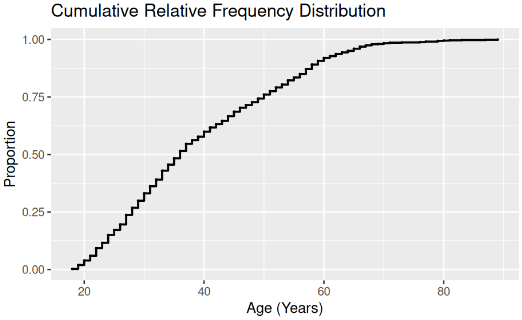
Range


Respondent Ages
Frequency
0
40
80
120
20
40
60
80
Years


Cumulative Relative Frequency Distribution
Proportion
0.00
0.25
0.50
0.75
1.00
20
40
60
80
Age (Years)

Interquartile and Semi-interquartile Range





Variance and Standard Deviation

$$D_i = X_i - \overline{X}$$

Variance and Standard Deviation

$$D_i = X_i - \overline{X}$$

$$S^2 = \frac{\sum_{i=1}^{N} D_i^2}{N} = \frac{\sum_{i=1}^{N} (X_i - \overline{X})^2}{N}$$

Variance and Standard Deviation

$$D_i = X_i - \overline{X}$$

$$S^2 = \frac{\sum_{i=1}^{N} D_i^2}{N} = \frac{\sum_{i=1}^{N} (X_i - \overline{X})^2}{N}$$

$$S = \sqrt{S^2} = \sqrt{\frac{\sum_{i=1}^{N} (X_i - \overline{X})^2}{N}}$$

Population vs. Sample Variation

$$S^2 = \frac{\sum_{i=1}^{N} D_i^2}{N} = \frac{\sum_{i=1}^{N} (X_i - \overline{X})^2}{N}$$

$$s^2 = \frac{\sum_{i=1}^{N} D_i^2}{N-1} = \frac{\sum_{i=1}^{N} (X_i - \overline{X})^2}{N-1}$$

Properties

- Range
  - Simple to compute
  - Unstable, very sensitive to outliers
- Interquartile Range
  - Less easy to compute
  - Robust
- Variance
  - Easy to compute
  - Sensitive to outliers
  - Unintuitive
- Standard deviation
  - Easy to compute
  - Intuitive